

The Real Body

Complex Analysis

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Definition

There is a set N equipped with an injective map $s : N \rightarrow N$, called the **successor function**[†], such that

- i) N has a unique element $\Theta \in N$ satisfying $s(n) \neq \Theta$ for all $n \in N$.
- ii) **(Principle of induction)** if A is a subset of N with the property that $\Theta \in A$ and that it is closed under s , then $A = N$.

The structure $(N, \Theta, s) \equiv \mathbb{N}$ is called a set of **natural numbers**.

[†]The image $s(n)$ for $n \in N$ is usually called the **successor of n** .

Definition

Let $(N, \Theta, s) \equiv \mathbb{N}$ and $(N', \Theta', s') \equiv \mathbb{N}'$ be two sets of natural numbers. An **isomorphism** is a bijection $\iota : N \rightarrow N'$ such that $\iota(\Theta) = \Theta'$ and $\iota(s(n)) = s'(\iota(n))$. If such a bijection exist, then we say that the two structures are **isomorphic**.

Theorem 1.1

If we define the relation $\sim$ by $\mathbb{N} \sim \mathbb{N}'$ if and only if $\mathbb{N}$ is isomorphic to $\mathbb{N}'$, then $\sim$ is an equivalence relation on the sets of natural numbers $\mathcal{N}$ and the set

$$\mathcal{N} / \sim$$

*is a singleton (this means that all the sets of natural numbers are isomorphic), whose unique element is called the set of **natural numbers**, denoted by $\mathbb{N}$.*

Definition

*The number Θ in $\mathbb{N}$ is known as **zero** and denoted by 0. The number $s(\Theta)$ is termed the number **one**, and denoted by 1.*

Theorem 1.2

There is a unique map $\alpha : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that

- i) $\alpha(n, 0) = n$ for all $n \in \mathbb{N}$;
- ii) $\alpha(n, 1) = s(n)$ for all $n \in \mathbb{N}$;
- iii) $\alpha(n, s(m)) = s(\alpha(n, m))$ for all $n, m \in \mathbb{N}$.

Definition

*The unique map α defined by the properties in the previous theorem is called the **addition map**. We usually write $\alpha(n, m) \equiv n + m$.*

Example 1.3

Let $s(1) \equiv 2$. Then show that $1 + 1 = 2$.

Theorem 1.4 (Commutativity of Addition)

If $n, m \in \mathbb{N}$, then $n + m = m + n$.

Theorem 1.5 (Associativity of Addition)

If $n, m, k \in \mathbb{N}$, then $(n + m) + k = n + (m + k)$.

Theorem 1.6 (Cancellation Law for Addition)

If $n, k \in \mathbb{N}$ then $n + k = n$ if and only if $k = 0$. Furthermore, for $m \in \mathbb{N}$ we have $n + k = m + k$ if and only if $n = m$.

Theorem 1.7 (Trichotomy Law in $\mathbb{N}$)

For any pair n, m of natural numbers, exactly one of the following statements hold true:

- a) $n = m$;
- b) $n + k = m$ for some $\mathbb{N} \ni k \neq 0$;
- c) $n = m + k$ for some $\mathbb{N} \ni k \neq 0$.

Definition

Given any pair n, m of distinct natural numbers, we say that n is **less than** [respectively, **bigger than**] m if the second [respectively, the third] statement on the previous theorem holds true. The relations $<, >$ on $\mathbb{N}$ are defined by

$$\begin{aligned}n < m &\iff n \text{ is less than } m, \\n > m &\iff n \text{ is bigger than } m.\end{aligned}$$

We further define $\leq, \geq$ by

$$\begin{aligned}n \leq m &\iff n < m \text{ or } n = m, \\n \geq m &\iff n > m \text{ or } n = m.\end{aligned}$$

Theorem 1.8

The relations less than and bigger than define strict partial orders on $\mathbb{N}$, on the other hand, the relations $\leq, \geq$ define partial orders on $\mathbb{N}$.

Theorem 1.9

There is a unique map $\mu : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that

- i) $\mu(n, 0) = 0$ for all $n \in \mathbb{N}$;
- ii) $\mu(n, 1) = n$ for all $n \in \mathbb{N}$;
- iii) $\mu(n, s(m)) = \mu(n, m) + n$ for all $n, m \in \mathbb{N}$.

Definition

*The unique map μ defined by the properties in the previous theorem is called the **multiplication map**. We usually write $\mu(n, m) \equiv nm$.*

Theorem 1.10 (Commutativity of Multiplication)

If $n, m \in \mathbb{N}$, then $nm = mn$.

Theorem 1.11 (Distributivity of Multiplication over Addition)

If $n, m, k \in \mathbb{N}$, then $n(m + k) = nm + nk$.

Theorem 1.12 (Associativity of Multiplication)

If $n, m, k \in \mathbb{N}$, then $(nm)k = n(mk)$.

Theorem 1.13 (Cancellation Law for Multiplication)

If $n, m, k \in \mathbb{N}$, then $m = k$ implies $nm = nk$. In particular, if also $n \neq 0$ then $nm = nk$ implies $m = k$.

Theorem 1.14

The relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ defined by

$$(m, n) \sim (k, l) \quad \text{if and only if} \quad m + l = n + k$$

is an equivalence relation.

Definition

*The set $\mathbb{Z}$ of all **integers** is defined as*

$$\mathbb{Z} = (\mathbb{N} \times \mathbb{N}) / \sim .$$

Example 1.15

The element $[(0, 1)]$ in $\mathbb{Z}$ is denoted by -1 . Similarly, we write $[(0, 2)] \equiv -2$.

Definition

The binary operations of **addition**, **subtraction** and **multiplication** on $\mathbb{Z}$, denoted by α' , σ , μ' , respectively, are defined via

$$\alpha'([(m, n)], [(k, l)]) = [(m + k, n + l)],$$

$$\sigma([(m, n)], [(k, l)]) = [(m + l, n + k)],$$

$$\mu'([(m, n)], [(k, l)]) = [(mk + nl, ml + nk)].$$

We usually write $\alpha'(a, b) = a + b$, $\sigma(a, b) = a - b$, $\mu'(a, b) = ab$ for any $a, b \in \mathbb{Z}$.

Theorem 1.16

The operations of addition, subtraction and multiplication are well-defined on $\mathbb{Z}$.

Theorem 1.17

The relation $\leq$ on $\mathbb{Z}$ defined by

$$[(m, n)] \leq [(k, l)] \iff m + l \leq n + k$$

is a partial order on $\mathbb{Z}$.

Definition

The canonical embedding $f : \mathbb{N} \hookrightarrow \mathbb{Z}$ is defined as

$$f(n) = [(n, 0)].$$

This map is a **homomorphism** in the sense that

$$f(m + n) = f(m) + f(n).$$

We usually write $\mathbb{N} \subseteq \mathbb{Z}$ to denote the fact that $f(\mathbb{N}) \subseteq \mathbb{Z}$.

Theorem 1.18 (Arithmetic properties of $\mathbb{Z}$)

For all $a, b, c \in \mathbb{Z}$ we have

- i) **(Commutativity for addition)** $a + b = b + a$.
- ii) **(Commutativity for multiplication)** $ab = ba$.
- iii) **(Associativity for addition)** $(a + b) + c = a + (b + c)$.
- iv) **(Associativity for multiplication)** $(a + b) + c = a + (b + c)$.
- v) **(Additive identity)** The element $0 = [(0, 0)] \in \mathbb{Z}$ is such that $a + 0 = a$ for each $a \in \mathbb{Z}$.
- vi) **(Multiplicative identity)** The element $1 = [(1, 0)] \in \mathbb{Z}$ is such that $a1 = a$ for each $a \in \mathbb{Z}$.
- vii) **(Additive inverse)** For each $a \in \mathbb{Z}$ there is a unique $b \in \mathbb{Z}$ such that $a + b = 0$. In this case we write $b \equiv -a$ and call b the **negative** of a .
- viii) **(Distributivity)** $a(b + c) = ab + ac$.

The latter properties are later used in abstract algebra to define a **commutative ring**.

Theorem 1.19

For any $a \in \mathbb{Z}$ we have $-a = (-1)a$.

Theorem 1.20

The relation $\sim$ on $\mathbb{Z} \times \mathbb{Z}$ defined by

$(a, b) \sim (c, d)$ if and only if $ad = bc$ and $b \neq 0, d \neq 0$, or $b = d = 0$,

is an equivalence relation.

Definition

The set $\mathbb{Q}$ of all **rational numbers** is defined as

$$\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z}) / \sim .$$

Theorem 1.21

The binary operations of **addition** and **multiplication** on $\mathbb{Q}$

$$[(a, b)] + [(c, d)] = [(ad + bc, bd)],$$

$$[(a, b)][(c, d)] = [(ac, bd)],$$

are well-defined. Further, we endow $\mathbb{Q}$ with a partial order $\leq$

$$[(a, b)] \leq [(c, d)] \quad \text{if and only if} \quad b, d \geq 0 \text{ and } ad \leq bc,$$

which is also well-defined.

Remark

The rational numbers contain an **additive identity** $0 = [(0', 1')]$ where $0', 1'$ are the additive identity and the multiplicative identity of $\mathbb{Z}$, respectively, and a **multiplicative identity** $1 = [(1', 1')]$.

Theorem 1.22 (Arithmetic Properties of $\mathbb{Q}$)

For all $a, b, c \in \mathbb{Q}$ we have

- i) **(Commutativity for addition)** $a + b = b + a$.
- ii) **(Commutativity for multiplication)** $ab = ba$.
- iii) **(Associativity for addition)** $(a + b) + c = a + (b + c)$.
- iv) **(Associativity for multiplication)** $(a + b) + c = a + (b + c)$.
- v) **(Additive identity)** The element $0 \in \mathbb{Q}$ is such that $a + 0 = a$ for each $a \in \mathbb{Q}$.
- vi) **(Multiplicative identity)** The element $1 \in \mathbb{Q}$ is such that $a1 = a$ for each $a \in \mathbb{Q}$.
- vii) **(Additive inverse)** For each $a \in \mathbb{Q}$ there is a unique $b \in \mathbb{Q}$ such that $a + b = 0$. In this case we write $b \equiv -a$ and call b the **negative** of a .
- viii) **(Multiplicative inverse)** For each nonzero element $a \in \mathbb{Q}$ there is a unique $b \in \mathbb{Q}$ such that $ab = 1$. In this case we write $b \equiv 1/a$ and call b the **reciprocal** of a .
- ix) **(Distributivity)** $a(b + c) = ab + ac$.

Remark

*All the properties from the previous theorem are later used in abstract algebra to define a **body** (or **algebraic field**), in particular, a totally ordered body when endowed with $\leq$.*

Theorem 1.23 (Archimedean Property)

For all $p, q \in \mathbb{Q}$ with $p > 0$, there exists a natural number $n \in \mathbb{N}$ such that $pn > q$.



Archimedes of
Syracuse
287 B.C.-212 B.C.

Historical Notes

Archimedes was the greatest mathematician of his age. His contributions in geometry revolutionised the subject and his methods anticipated the integral calculus. He was a practical man who invented a wide variety of machines including pulleys and the Archimidean screw pumping device. His father was Phidias, an astronomer. We know nothing else about Phidias other than this one fact and we only know this since Archimedes gives us this information in one of his works. Archimedes was a native of Syracuse, Sicily. It is reported by some authors that he visited Egypt and there invented a device now known as Archimedes' screw. This is a pump, still used in many parts of the world. It is highly likely that, when he was a young man, Archimedes studied with the successors of Euclid in Alexandria. Certainly he was completely familiar with the mathematics developed there, but what makes this conjecture much more certain, he knew personally the mathematicians working there and he sent his results to Alexandria with personal messages. He regarded Conon of Samos, one of the mathematicians at Alexandria, both very highly for his abilities as a mathematician and he also regarded him as a close friend.

Definition

A sequence in $\mathbb{Q}$ is a function $X : \mathbb{N} \rightarrow \mathbb{Q}$, whose image at some particular n is often written as x_n and whose image $X(\mathbb{N})$ is sometimes denoted by (x_n) .

Definition

Let S be a nonempty subset of a totally ordered field $\mathbb{R}$.

- i) The set S is said to be **bounded above** if there exists a number $u \in \mathbb{R}$ such that $s \leq u$ for all $s \in S$. Each number u is called an **upper bound** of S .
- ii) The set S is said to be **bounded below** if there exists a number $w \in \mathbb{R}$ such that $w \leq s$ for all $s \in S$. Each number w is called a **lower bound** of S .
- iii) A set is said to be **bounded** if it is both bounded above and bounded below. A set is said to be **unbounded** if it is not bounded.

Definition

Let S be a nonempty subset of a totally ordered field $\mathbb{R}$.

- i) If S is bounded above, then a number u is said to be a **supremum** (or a **least upper bound**) of S if it satisfies the conditions:
 - a) u is an upper bound of S , and
 - b) if v is any upper bound of S , then $u \leq v$.
- ii) If S is bounded below, then a number w is said to be an **infimum** (or a **greatest lower bound**) of S if it satisfies the conditions:
 - a') w is a lower bound of S , and
 - b') if t is any lower bound of S , then $t \leq w$.

If the supremum or the infimum of a set S exists, we will denote them by

$$\sup S \quad \text{and} \quad \inf S.$$

Definition

A sequence X in a totally ordered field $\mathbb{R}$ equipped with a metric is said to **converge** if there is some $x \in \mathbb{R}$ such that given any $\varepsilon > 0$ there is a natural number $K \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for any $n \geq K$. Furthermore, X is said to be **increasing** if

$$x_1 \leq x_2 \leq \cdots \leq x_n \leq x_{n+1} \leq \cdots ,$$

and **decreasing** if

$$x_1 \geq x_2 \geq \cdots \geq x_n \geq x_{n+1} \geq \cdots .$$

A sequence is said to be **monotone** if it is either increasing or decreasing.

Definition

We say that a sequence of intervals (I_n) is **nested** if the following chain of inclusions holds:

$$I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq I_{n+1} \supseteq \cdots .$$

Definition

Let $X = (x_n)$ be a sequence in a totally ordered field $\mathbb{R}$ and let $n_1 < n_2 < \dots < n_k < \dots$ be a strictly increasing sequence of natural numbers. Then the sequence $X' = (x_{n_k})$ is called a **subsequence** of X .

Definition

The **absolute value** is a map $|\cdot| : \mathbb{Q} \rightarrow \mathbb{Q}$ defined as

$$|x| = \begin{cases} x & x \geq 0, \\ -x & x < 0. \end{cases}$$

Definition

A sequence in $\mathbb{Q}$ is said to be **Cauchy** (or to be a **Cauchy sequence**) if for each rational number $\varepsilon > 0$ there is a natural number $H \in \mathbb{Q}$ such that

$$|x_n - x_m| < \varepsilon \quad \text{for } n, m \geq H.$$

Theorem 1.24

Let $\mathbb{R}$ denote an arbitrary totally ordered field. Then the following are equivalent:

- i) **(Least upper bound property)** Every nonempty subset $S \subseteq \mathbb{R}$ bounded from above has a supremum.
- ii) **(Monotone convergence theorem)** Every bounded monotone sequence in $\mathbb{R}$ is convergent.
- iii) **(Nested intervals property)** Every nested sequence of closed and bounded intervals (I_n) in $\mathbb{R}$ has a nonempty intersection

$$\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset.$$

- iv) **(Bolzano-Weierstrass theorem)** Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
- v) **(Cauchy criterion)** A sequence in $\mathbb{R}$ is Cauchy if and only if it is convergent.

Definition

*A totally ordered field $\mathbb{R}$ is said to be **complete** if any (and thus every) of the properties in the previous theorem hold true.*

Theorem 1.25 (Babylonian Method for calculating Square Roots)

Let $a > 0$ be a rational number and take any other $s_1 > 0$, and thus define

$$s_{n+1} = \frac{1}{2} \left(s_n + \frac{a}{s_n} \right).$$

The sequence $S = (s_n)$ as defined above converges to $\sqrt{a}$ (i.e., to a number x with the property that $xx = a$).

Theorem 1.26 ($\sqrt{2} \notin \mathbb{Q}$)

There does not exist a number $r \in \mathbb{Q}$ such that $rr = 2$.

Example 1.27

The set $\mathbb{Q}$ of all rational numbers is not complete. In particular, the sequence $X = (x_n)$ in $\mathbb{Q}$ defined by

$$x_1 = 1, \quad x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$$

is a Cauchy sequence that converges to a number x with the property that $xx = 2$ (by Theorem 1.25) and hence it does not converge in $\mathbb{Q}$ (by Theorem 1.26).

Theorem 1.28

Let $\mathbf{C}$ be the collection of all Cauchy sequences in $\mathbb{Q}$. Then the relation $\sim$ defined via $X \sim Y$ if and only if given any rational $\varepsilon > 0$ there is a natural number $K \in \mathbb{N}$ such that

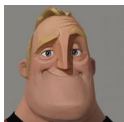
$$|x_n - y_n| < \varepsilon \quad \text{for } n \geq K,$$

is an equivalence relation.

Definition

*The set $\mathbb{R}$ of all **real numbers** is defined as*

$$\mathbb{R} = \mathbb{C} / \sim .$$



Counting numbers, haha

$$\mathbb{N} = \{1, 2, 3, \dots\}$$



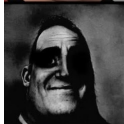
Just two natural numbers!

$$\mathbb{Z} = \{a - b \mid a, b \in \mathbb{N}\}$$



Just two integers!

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$$



**The limits of infinite convergent
Cauchy sequences of rational numbers**

$$\mathbb{R} = \left\{ \lim_{n \rightarrow \infty} a(n) \mid a : \mathbb{N} \rightarrow \mathbb{Q}, \forall \varepsilon > 0. \exists N \in \mathbb{N}. \forall n > N. |a(n) - a(N)| < \varepsilon \right\}$$



Just two real numbers!

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

Figure 1: Complex analysis meme

Theorem 1.29

The operations of **addition**, **multiplication** and the partial order $\leq$ on $\mathbb{R}$

$$[(x_n)] + [(y_n)] = [(x_n + y_n)],$$

$$[(x_n)][(y_n)] = [(x_n y_n)],$$

$$[(x_n)] \leq [(y_n)] \iff \left([(x_n)] = [(y_n)] \text{ or } \exists K \in \mathbb{N} : x_n < y_n \text{ for } n \geq K \right),$$

are well-defined.

Theorem 1.30

Let $\mathbb{R} \ni 0 = [(0', 0', \dots)]$ where $0'$ is the additive identity of $\mathbb{Q}$. Then for all $x \in \mathbb{R} \setminus \{0\}$ there exists $\mathbb{Q} \ni \varepsilon > 0$ such that exactly one of the following holds for all representatives (x_n) of x in $\mathbb{C}$:

- i) $x_n > \varepsilon$ for all but finitely many $n \in \mathbb{N}$.
- ii) $x_n < -\varepsilon$ for all but finitely many $n \in \mathbb{N}$.

In particular, for any $x \in \mathbb{R} \setminus \{0\}$ we can obtain a sequence (p_n) from any representative (x_n) of x by discarding finitely many terms of (x_n) , with the property that either $p_n > \varepsilon$ for all $n \in \mathbb{N}$ or else $p_n < -\varepsilon$ for all $n \in \mathbb{N}$.

Theorem 1.31 (Arithmetic Properties of $\mathbb{R}$)

The following statements hold true:

- i) *The pairs $(\mathbb{R}, +)$ and $(\mathbb{R} \setminus \{0\}, \cdot)$ are Abelian groups.*
- ii) *$(\mathbb{R}, \leq)$ is a totally ordered set.*
- iii) *The field operations are **compatible** with the partial order $\leq$ in the sense that for all $x, y, z \in \mathbb{R}$ we have*
 - a) **(Order preservation under addition)** *If $x \leq y$, then $x + z \leq y + z$.*
 - b) **(Order preservation under multiplication)** *If $0 \leq x$ and $0 \leq y$, then $0 \leq xy$.*

Remark

This construction of $\mathbb{R}$ is complete by construction, in particular, every Cauchy sequence is convergent (the converse always holds true).

Theorem 1.32 (Uniqueness of $\mathbb{R}$)

Every complete totally ordered field is both isomorphic and order-isomorphic to $\mathbb{R}$, so in this sense all constructions of $\mathbb{R}$ are equivalent.